# State-Space Lab 3:<br>Ball-Balancing-Board

## 1 System Description

The plant to be controlled in this lab (see Figure 1) consists of a ball that can roll freely on a platform (“balancing board”) which can tilt in two dimensions.

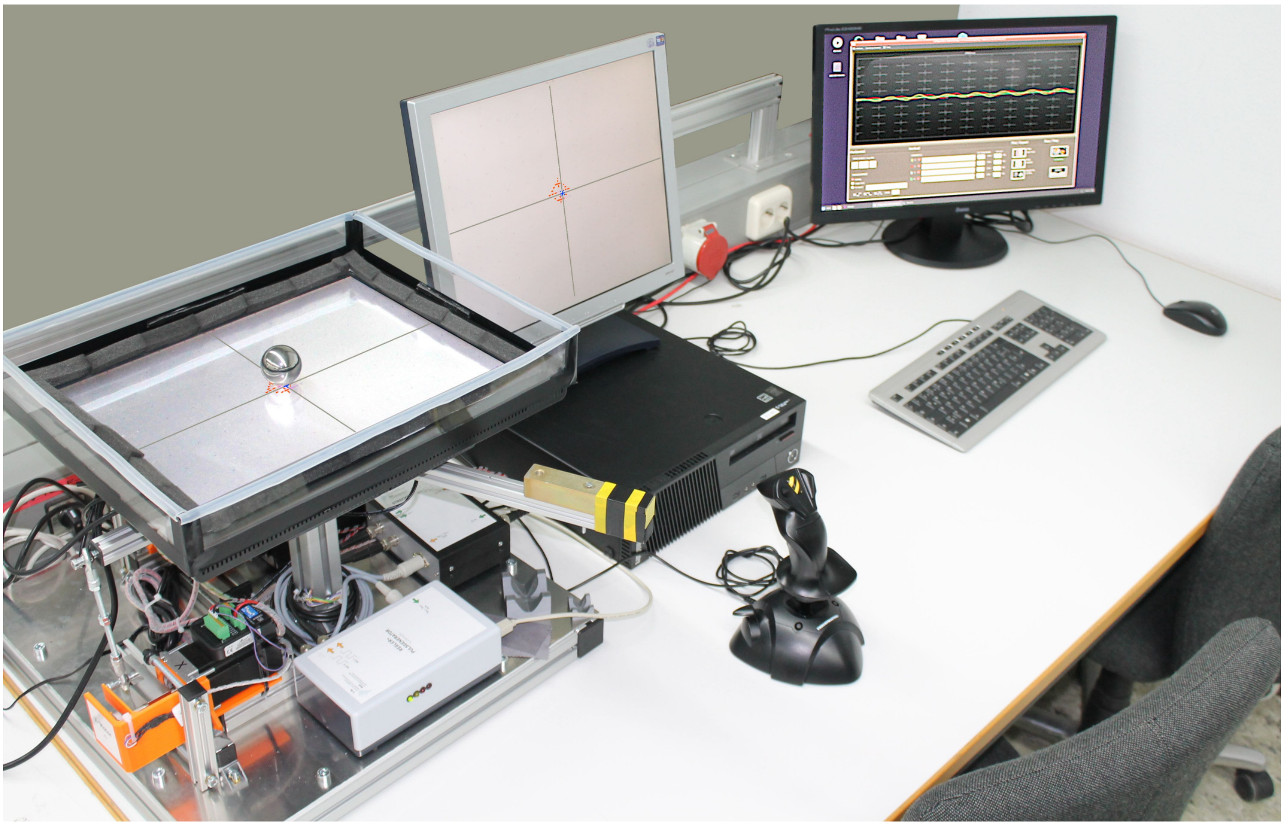

*Figure 1: The plant to be controlled (the ball balancing board with the ball at the center of the platform) is shown on the left in the figure, and on the right is the control PC with GUI for measurement and control.*

The platform is centrally mounted on a universal joint (cardan joint). Two servo motors act as cranks (one for each dimension x and y) and can tilt the platform via connecting rods.
The crank angles $\alpha=[\alpha_x,\alpha_y]$ can be specified. This results in the platform tilt angles $\beta=[\beta_x,\beta_y]$ as shown in Figure 2. The ball position $p_B=[x_B,y_B]$ on the platform is to be controlled.

Regarding the kinematics, the following simplifying assumptions are made:

- The platform’s pivot point lies in the plane of motion of the ball (no vertical offset).
- The crank angles $\alpha$ are small, such that $\sin(\alpha)\approx\alpha$.
- There is no mechanical backlash.

Under these assumptions, there is a proportional relationship between the crank angles $\alpha$ and the platform tilt angles $\beta$.

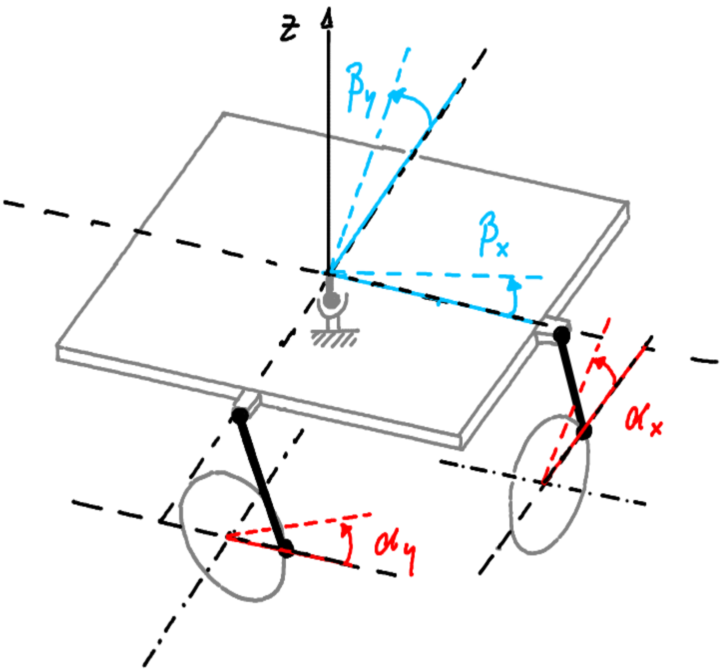


*Figure 2: Three-dimensional sketch of the plant's geometry and kinematics showing the platform centrally mounted on the cardan joint with its tilt angles β, as well as the two motors with their crank angles α connected to the platform via rods.*

Regarding the dynamics, the following simplifying assumptions are made:

- There is no friction between the moving parts.
- Aside from its own weight, no other significant forces act on the ball. As the tilt angle $\beta$ increases, the force $F$ at the ball's contact point increases. This generates a torque about the ball's center resulting in the rolling motion of the ball (see Figure 3).

Hence, for small tilt angles $\beta$, the angular acceleration of the ball about its center can also be assumed to be proportional to the tilt angle $\beta$.

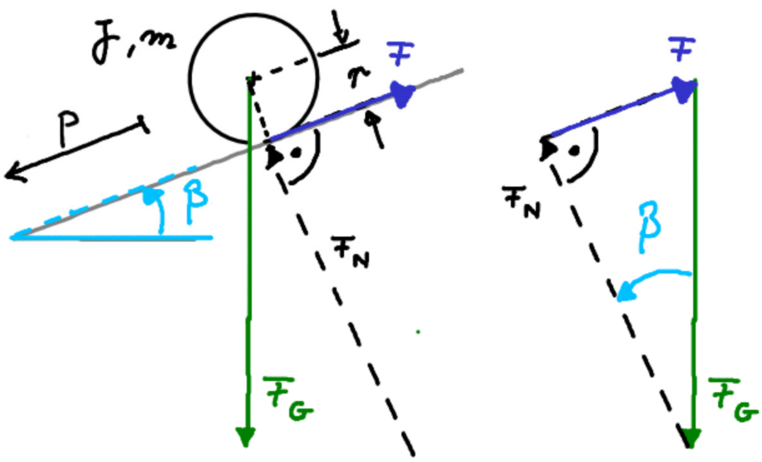


*Figure 3: Ball rolling on the tilted plane along dimension p (only one dimension is shown).*
*m: mass of the ball, J: moment of inertia of the ball, r: radius of the ball*
*$F_G$: gravitational force of the ball, $F_N$: resulting normal force, F: resulting force at the ball's contact point*

# 2 Preparation Exercises (to be completed in advance of the lab session!)

## 2.1 Identification of the dynamic behavior of the subsystems

a) The measured step response of one of the position-controlled motors is shown in Figure 4.

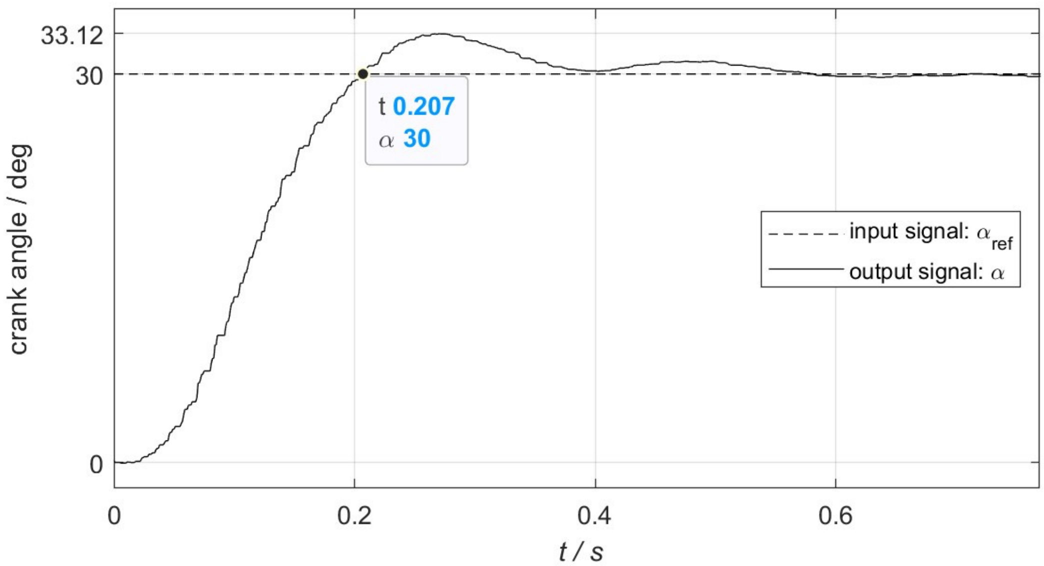


*Figure 4: Step response of the position-controlled motor*

Approximate the system behavior by a $PT_2$ system with the transfer function

$$G_M(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K_M}{T_0^2 s^2 + 2\zeta T_0 s + 1} .$$

Use the characteristic values (final value, overshoot, rise time) of the given step response to determine the transfer function parameters

- static gain $K_M$,
- damping ratio $\zeta$ and
- time constant $T_0$ (reciprocal of the natural angular frequency $\omega_n$)

It is assumed that the two servo motors exhibit the same dynamic behavior.

b) For simplicity, it is assumed that the acceleration of the ball is proportional to the crank angle $\alpha$ (measured in degrees) of the motor that tilts the inclined plane by a tilt angle $\beta$:

$$\ddot{p}_B(t) = K_B \cdot \alpha(t)$$

The proportional gain $K_B$ can be determined from the acceleration of a homogeneous sphere rolling down an inclined plane under pure rolling conditions (without slipping). Research the corresponding analytical formula in the literature and apply it to this problem.
Given that a crank angle of $\alpha = 15°$ results in a plane tilt angle of $\beta = 3.5°$, determine the numerical value of $K_B$ in units of $m/(s^2 \cdot deg)$.

It is assumed that the ball exhibits the same dynamic behavior in x- and y-direction.

### 2.2 Simulink model of the plant

a) Model the position-controlled servo motor in Simulink using a transfer function block. Add a saturation block to limit the crank angle (upper limit 45°, lower limit −45°).

b) Model the ball-on-board behavior in Simulink using a gain for $K_B$ and two integrators.

c) The input variable $u(t)$ of the plant is the reference crank angle $\alpha_{ref}(t)$ of the motor. Use a "From Workspace" block to read the input signal $u(t)$ from the MATLAB workspace into the Simulink model.

d) Extend the model by derivative blocks to obtain the crank angular velocity $v_\alpha(t)$ from the motor crank angle $\alpha(t)$ and to obtain the ball velocity $v_B(t)$ from the ball position $p_B(t)$.

e) Use "To Workspace" blocks to export the following signals to the MATLAB workspace for further analysis:
   $p_B(t)$: ball position (unit: m)
   $v_B(t)$: ball velocity (unit: m/s)
   $\alpha(t)$: motor crank angle (unit: deg)
   $v_\alpha(t)$: motor crank angular velocity (unit: deg/s)

f) Extend the model to allow disturbances to be applied to the crank angle, to the ball velocity and to the ball position.

### 2.3 State-space model of the plant

a) Determine the second-order differential equation of the servo motor from the PT$_2$ the transfer function with symbolic parameters $K_M$, $\zeta$ and $T_0$:

$$G_M(s) = \frac{\alpha(s)}{\alpha_{ref}(s)} = \frac{K_M}{T_0^2 s^2 + 2\zeta T_0 s + 1} .$$

b) Starting from the two second-order differential equations describing the servo motor and the ball-on-board dynamics, derive the corresponding state-space model of the plant.
Use the symbolic system parameters $K_B$, $K_M$, $\zeta$ and $T_0$ and the following state definition:
   $x_1(t)$: ball position $p_B(t)$
   $x_2(t)$: ball velocity $v_B(t)$
   $x_3(t)$: crank angle $\alpha(t)$ of the motor
   $x_4(t)$: crank angular velocity $v_\alpha(t)$ of the motor

The input variable of the plant is the reference crank angle $\alpha_{ref}(t)$. The output variable is the ball position $p_B(t)$.

### 2.4 Analysis of the plant in MATLAB/Simulink

a) Implement the state-space model of the plant in MATLAB.

b) Determine the eigenvalues of the system. Is the system stable?

c) Analyze the controllability and observability of the system.

d) Use your Simulink model of the plant to simulate the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$ to a step input in the reference crank angle $u(t)=\alpha_{ref}(t)$, changing from 0 to 10°.

Plot the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$ and the input signal $u(t)$ in separate subplots within a single figure. Add axis labels, meaningful titles and legends (not just $x_1$, $x_2$…).
All plots should be displayed using the same time duration and time-axis scaling. Ensure that the data is sampled with sufficiently high resolution.

# 3 Exercises for the Lab Session

## 3.1 Model of the PI state-feedback control system

a) Derive the extended state-space model for the PI state-feedback design and implement it in MATLAB.
Note: The MATLAB model is used for the state-feedback design only, not for the simulation.

b) Extend your Simulink model of the plant developed in Section 2.2 as follows:
(1) Implement a state-feedback controller. Use four separate scalar "Gain" blocks, one for each state variable, for state feedback.
(2) Extend the system to a PI state-feedback control scheme.
(3) Use a "From Workspace" block to read the reference signal w(t) from the MATLAB workspace into the Simulink model.

## 3.2 Design of a PI state-feedback controller by pole placement

a) Determine the desired closed-loop pole locations to achieve a closed-loop system response with a settling time of $t_{s,98\%}=4$ seconds and no overshoot.
Place the remaining poles five times further to the left than the dominant poles.
Determine the state-feedback gains of the extended system and derive the corresponding feedback gains for the original plant states and the PI controller gains. Set $K_P=1$.

b) Simulate the state responses of the PI state-feedback control system to the following reference input $w(t)$ and disturbances:
$t=0$: step change in the reference input $w(t)$ from 0 to 0.2m
$t=15$: disturbance step from 0 to 1° in the crank angle
$t=30$: disturbance step from 0 to 0.1m/s in the ball velocity
$t=45$: disturbance step from 0 to 0.05m in the ball position

Plot the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$ and the actuating signal $u(t)$ in separate subplots within a single figure. Add axis labels, meaningful titles and legends (not just $x_1$, $x_2$…).
All plots should be displayed using the same time duration and time-axis scaling. Ensure that the data is sampled with sufficiently high resolution.

c) Show your simulation results to your supervisor.
Afterwards, test your designed PI state-feedback controller on the real system
- for a fixed setpoint of the ball position.
- for a reference trajectory of the ball.
Apply different disturbances to the system in both cases.
Document and comment your results.

d) Repeat steps a)-c) with modified design requirements:
Settling time $t_{s,98\%}=4$ seconds and overshoot approx. 4-5%. Set $K_P=1$.

### 3.3 Design of a PI state-feedback controller by LQR

a) Design an optimal PI state-feedback controller based on the weighting matrices

$$Q=\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0.01 & 0 \\ 0 & 0 & 0 & 0 & 1000 \end{pmatrix} \qquad R=r=1$$

Determine the solution P of the algebraic Riccati equation, the state-feedback gains of the extended system and the locations of the closed-loop poles.
Derive the corresponding feedback gains for the original plant states and the PI controller gains. Set $K_P=1$.

b) Simulate the state responses of the PI state-feedback control system.
Use the same reference input w(t) and disturbances z(t) as given in exercise 3.2.
Plot the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$ and the actuating signal $u(t)$ in separate subplots within a single figure. Add axis labels, meaningful titles and legends (not just $x_1$, $x_2$…).
All plots should be displayed using the same time duration and time-axis scaling. Ensure that the data is sampled with sufficiently high resolution.
Document and comment your results. Compare the results with the results of 3.2.

c) Show your simulation results to your supervisor.
Afterwards, test your designed PI state-feedback controller on the real system
- for a fixed setpoint of the ball position.
- for a reference trajectory of the ball.

Apply different disturbances to the system in both cases.
Document and comment your results and compare them with exercise 3.2.

d) Repeat steps a)-c) with modified design requirements:
Reduce the weighting factor $Q(5,5)$, which corresponds to the integral of the control error in the $Q$ matrix. All other specifications remain unchanged.
Explain how this change affects the closed-loop behavior and justify the observed effects.

e) Discuss further cases with your supervisor.